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RPC/001/021 – Management Process for Defect Investigations of Network Plant, Equipment or Materials

1. Purpose

The purpose of this document is to define the management process for the reporting, investigation and completing of corrective actions for defects or failures of products operating on the distribution networks of CE Electric UK.

Where necessary the process may result in a corrective action plan that eliminates the cause of similar defects. Results of corrective actions shall be reviewed and analysed for effectiveness. An investigation may establish learning points related to the product under investigation and result in a change of product design, working practices, training requirements, investment policy etc. The process should also identify where a product has failed prematurely and result in warranty discussions with the supplier.

This document identifies the roles and responsibilities of each section during the process. Guidance is given on the initiators that identify an investigation requirement and how they are prioritised.

2. Scope

This code of practice covers all products on or intended for use on Low Voltage (LV), High Voltage (HV) and Extra High Voltage (EHV) networks of CE Electric UK and shall be followed by all employees and contractors operating on behalf of CE Electric UK.

The code of practice sets out the process to be followed in the event of a product being identified as failing to meet the required function(s), performance and condition.

3. Management Process

3.1 General

All network products which fail to meet the required function(s), performance and condition shall be reported and investigated in accordance with this process. Defects resulting in an accident, incident or dangerous occurrence shall follow CE Electric UK policy HAS/011 – Accident Reporting and Investigations. The use of the CE Electric UK “159 field reporting telephone hotline” shall initially be used to report details and seek advice.

The collation of defective or failed product information will allow analysis of trends, supplier quality performance, training or process requirements and product performance whilst identifying the cost of defects to CE Electric UK.

The process shall be coordinated by the Defect Process Co-ordinator who will be a Standards and Assessments Engineer nominated by the Standards and Assessments Manager. Internal support shall be provided if requested and external resource utilised where necessary. The management of the process shall be carried out in a timely and cost effective manner. The process shall follow the flow chart detailed in Appendix 2.

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3.2 Roles and Responsibilities

This section details the roles and responsibilities of all persons that may be involved in the defect reporting and subsequent investigation process.

Industrial, Supervising, Warehouse and Engineering Staff (including contractors working on behalf of CE Electric UK) shall be responsible for: -

- The reporting of any products found to be defective or damaged.
- The completion of defect forms where required.
- Providing field and product information to Control Engineers, Restoration Dispatchers and the Defect Process Co-ordinator where required.
- The returning of faulty products to a location agreed with the Defect Process Co-ordinator.

Network Control Engineers shall be responsible for: -

- The input of HV and EHV product defect information into IRIS.
- Providing background event information to the Defect Process Co-ordinator, Defect Investigation Co-ordinator or Defect Panel Members.

Restoration Dispatchers shall be responsible for: -

- The input of LV product defect information into IRIS.

Defect Process Co-ordinator is the Standards and Assessments Engineer nominated by the Standards and Assessments Manager and shall be responsible for: -

- Reviewing defect forms and fault reports including IRIS.
- Recording reported defects.
- Providing guidance, information and feedback to persons reporting defects.
- Deciding the type of investigation required.
- Collating, retaining and storage of all relevant information.
- Update of appropriate records.

Defect Investigation Co-ordinator will normally be a member of the Standards and Assessments Section but occasionally may be from another specialist section (i.e. PEP or Protection staff etc). This appointment shall be agreed with the Defect Investigation Co-ordinator and Standards Resource Co-ordinator if necessary. They shall be responsible for: -

- The formation of an investigation group with appropriate representation.
- The effective management of the investigation
- Co-ordinating external resource if required.
- Producing Investigation Summary Papers.
- Providing and monitoring actions issued to investigation group members, CE Electric UK staff or external resource.
- Ensuring that corrective action, inspection or remedial programs are accepted by the appropriate section of CE Electric UK.

Major Defect Panel Members will normally be nominated by the Defect Investigation Co-ordinator or their section managers following involvement with the defect. Alternatively they may be a specialist for that particular product type and shall be responsible for: -

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- Providing support and carrying out actions issued during the defect investigation process.
- As required liaise with customers impacted by the defect.
- Providing sign-off of Investigation Summary papers where required.

3.3 Products Covered

Distribution network products at all voltages, which have been purchased by CE Electric UK or contractors working on their behalf, are included in this process. Appendix 5 details specific product types.

The following defect failure modes will be recorded and/or investigated: -

- Defective Products – discovered prior to or post installation.
- Failed Products – during or post installation.
- Products not conforming to specification.
- Incorrect labelling (including data plate, product reference etc).
- Stock products with shelf life expired.

3.4 Selection Criteria

The following guidance/triggers shall be followed by all staff and contractors when deciding if a defect/failure is to be reported. A defect shall be reported if any of the following criteria are relevant: -

- All EHV plant, cables, joints, terminations and overhead line components
- New product prior to or during commissioning
- A defective product less than 10 years old
- From other evidence it can be assumed that the product may be less than 10 years old (see **note*** below).
- Any product deemed to be exhibiting an adverse failure trend.
- Intelligence that a particular product, of any age, has a history of unpredicted failure.
- Restrictions on the defective product would have a severe impact of customers of CE Electric UK.

Note * For example a product could be less than 10 years old if: -

- A cable joint is cold applied
- The cable into a joint is XLPE
- The overhead line insulator is silicon rubber
- The pole mounted plant has silicon rubber insulators (ABSD, expulsion fuse sw, surge arrester etc)
- An ABSD is high level operated
- LV linkboxes are of a modular design
- LV network is ABC
- HV conductors are XLPE insulated

*If there is a doubt for the age of the product it should be reported.

3.5 Reporting

The reporting of defects and failures shall be carried out at the first available opportunity and submitted to the Defect Process Co-ordinator, verbally via Control Engineers and Restoration Dispatchers and/or on CE Electric UK Defect Report Forms. The person reporting the defect will be kept informed at key stages during the investigation and of the investigation outcome.

3.5.1 EHV and HV Defects

For EHV and HV failures the Control Engineer shall be responsible for collating all the required information from operational staff on site. The detail shall be submitted to Network Control Support on the standard IRIS template. The following information shall be provided if available: -

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- Type of product that has failed.
- Size and/or rating.
- Voltage.
- Manufacturer.
- Serial number.
- Location.
- Brief details of the nature of the defect.
- Date/time of incident.
- Name of operational staff on site.
- Details of consequences of the failure (safety issues, other apparatus effected etc.).
- Any additional background information of relevance.

Minor defects of EHV and HV apparatus (e.g. cosmetic damage, quality issues) shall be reported on a Defect Report Form, as shown in Appendix 1 and submitted to the Defect Process Co-ordinator.

DIN's, SOP's and NEDeRS

The reporting and receiving of Dangerous Incident Notifications (DINs), Suspension of Operational Practices (SOPs) and National Equipment Defect Reporting Scheme (NEDeRS) is not affected by this management process and shall continue to follow the current reporting process within CE Electric UK. However, any associated investigation shall follow the process outlined in this document.

3.5.2 LV Defects

Product failures on the LV network shall be reported direct to the Defect Process Co-ordinator or via the Restoration Dispatcher for input into the IRIS reporting system. The following information shall be provided if available: -

- Type of product that has failed.
- Size and/or rating.
- Voltage.
- Manufacturer.
- Serial number.
- Location
- Brief details of the nature of the defect.
- Date/time of incident.
- Name of operational staff on site.
- Details of consequences of the failure (safety issues, other apparatus effected etc.).
- Any additional background information of relevance.

Minor defects of LV apparatus (e.g. cosmetic damage, quality issues) shall be reported on a Defect Report Form, as shown in Appendix 1 and submitted to the Defect Process Co-ordinator.

3.6 Preservation of Defective/Failed Product

In general if a defective product is to be investigated, it shall be returned complete with all component parts and packaging to an agreed location. It shall be packaged and transported in such a way to preserve the contents. The Defect Process Co-ordinator will provide guidance on where defective products shall be returned.

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3.6.1 Catastrophic Plant Failure

Defects of this type shall follow CE Electric UK policy HAS/011 – Accident Reporting and Investigations. Site safety shall be maintained at all times and it must also be ensured that all assets are protected against further damage and/or the contamination of evidence where possible. The following shall be considered: -

- The presence of pressurised components
- Contamination (SF6, PCB and Oil etc)
- Structural integrity of the building
- Which equipment might be live
- The continuity of earthing systems

The quality, continuity and restoration of the customers supply shall also be considered as a priority.

It is important that once made safe, the defective product remains, where possible, undisturbed until photographic evidence is taken of the product and surrounding area.

Identification labels shall be added to the photographs and once on site investigation has been completed, the product shall be carefully handled, stored and transported to minimise any further damage.

3.6.2 Joint, Termination and Cable Failures

Joints and Terminations

If a cable joint or termination failure is to be investigated the following is required: -

- The joint including approximately 500mm of each cable entering and leaving the joint.
- The termination including approximately 500mm of the cable entering the termination.
- Each cable shall have an end cap applied to prevent the ingress of moisture.
- Any packaging that details product information.

Cable and Conductor

If a cable failure is to be investigated the ends shall also be fitted with an end cap. Drum numbers shall be provided for defective cable or conductor if known.

Details of any backfill material should also be noted if it is possible that it may be a contributing factor.

3.7 Investigation Process

For all reported defects the Defect Process Co-ordinator shall be responsible for the collation of all relevant information and input onto the defect register. The Defect Process Co-ordinator is responsible for deciding the course of action based on the requirements of Section 3.4 of this document. The process shall follow the basic flow chart detailed in Appendix 2.

3.7.1 Defect Register

The defect register shall contain the basic information necessary to provide a high level overview of the defect(s) and shall be updated to reflect investigation progress. The register shall be available for review by System Strategy to analyse and monitor trends.

3.7.2 Investigation Priority

General guidelines for the prioritisation of investigations are detailed in the following categories below: -
Priority 1

- Any equipment defect that has caused injury (or had the potential to), a major environmental hazard or significant interruption to supplies.

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- Equipment that has suffered catastrophic or disruptive mechanical or electrical failure resulting in a fire, explosion or damage.
- Any equipment defects that pose (or have the potential to pose) a significant risk to:
 - The safety of staff or public.
 - System security and/or unacceptable network performance.
 - Environmental breach.
 - Statutory breach.
 - A significant financial impact to the business.

Priority 2

- All equipment that is reported as defective or damaged prior to installation and that is of a minor nature.
- Products that do not conform to the purchased specification.
- Products that are incorrectly labelled or have component parts missing.

3.7.3 Types of Investigation

Recording for Information Only

Where a product defect that has previously been investigated and the outcome accepted, then any future defects of the same nature and product type, shall be recorded on the defect register for information purposes.

Basic Investigation

For minor defects the Defect Process Co-ordinator will normally adopt the role of Defect Investigation Co-ordinator and carry out the investigation. A basic investigation shall be undertaken on: -

- Priority 2 defects as defined in section 3.7.2.
- Damaged or defective products where minimum technical product knowledge is required.

A basic investigation process will result in the completion of the Defect Investigation Summary Paper as detailed in Appendix 3. This shall be authorised at an appropriate level and retained. The person reporting the defect and any other appropriate parties shall be informed of the outcome. The Defect Process Co-ordinator is responsible for ensuring that any corrective actions have been completed and changes to procedures and/or documentation have been completed.

Major Investigation

Major defect investigations shall be managed by the Defect Investigation Co-ordinator and will require the formation of an investigation team. The investigation team will include representation from the relevant sections of CE Electric UK and/or external resource where required. As a minimum, internal representation shall be provided by Network Operations, System Strategy, Standards and Assessment, and Programme Delivery. Major investigations shall include: -

- Priority 1 defects as defined in section 3.7.2.
- Defective products where a high level of technical knowledge is required.

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When carrying out a major investigation a number of contributing factors shall be considered. They include lightning activity, system disturbance traces, local switching and auto-recloser or testing activity. Copies of inspection and maintenance records may also provide an insight into the failure mode. The investigation process will result in the completion of the Defect Investigation Summary Paper with the content agreed and signed off by the investigation team. This paper shall be authorised at an appropriate level and retained by the Defect Process Co-ordinator. The person reporting the defect and any other appropriate parties shall be informed of the outcome. The Defect Process Co-ordinator is responsible for ensuring that any actions detailed in Section 3.8.2 of this document have been completed after the investigation.

3.7.4 Investigation Considerations

When conducting a defect investigation the following shall be considered: -

- Safety implications
- Operational restriction impact
- Site access restrictions
- Environmental implications
- Implication on network performance
- Customer service impact
- Work load implications including the requirement for a maintenance, replacement or retro-fit program(s)
- Financial implications (total life cycle costs)
- Defects of a similar mode with other products

During the investigation process a decision may be required with regard to whether a failed unit is to be refurbished or replaced. This decision will be agreed by the working group based on the CE Electric UK asset replacement policy.

3.7.5 Communications

For technical communications a nominated person from the defect group or the Defect Investigation Co-ordinator may be required to liaise with the product supplier, external resource or customers of CE Electric UK impacted by defect. Commercial communications shall only be carried out by the Purchasing Section under the guidance of the Purchasing Operations Manager.

All information collated during the investigation shall be treated as confidential and not released to third parties without written permission from an appropriate CE Electric UK authority.

3.7.6 Investigation Folder Contents

The Defect Investigation Co-ordinator is responsible for creating a specific folder for each investigation and collating all the relevant information. This shall include the Defect Report Form, Defect Summary Paper, Event Log (as shown in Appendix 4), any communications, external reports and any other relevant information. Signed copies of Defect Summary Papers shall also be retained by the defect co-ordinator in the investigation folder.

3.8 Investigation Outputs

3.8.1 Investigation Outcome

The results from an investigation may result in one or more of the following options being documented as the root cause of the defect: -

- The product was defective when supplied.

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- The product supplied was not as specified by CE Electric UK.
- Staff or contracting staff did not install the product as documented in the business or suppliers work instructions.
- Staff or contracting staff used the product outside of the original design parameters.
- The product had suffered damaged during storage, transport to site, erection, connection or commissioning etc.
- The product failed due to the network configuration or system parameters.
- The product reached the end of its useful life due to normal aging and/or wear.
- The product failed due to third party damage/interference.

If it is found that employees have been in breach of company specification, procedures or policy requirements disciplinary action shall be carried out in accordance with CE Electric UK document HRP/005 – Code of Practice and Business Ethics.

3.8.2 Investigation Actions

The outcome of the investigation process may result in a number of required actions including: -

- Compensation discussions with suppliers via the Purchasing Section
- Recall of defective or potentially defective products
- Repair of the defective product
- Reporting of DIN's, SOPS etc.
- Arranging of replacement products via the Logistics Section
- Placing/lifting of operational restrictions
- Changes to the CE Electric UK technical specification by the Standards and Assessments Section
- Assessment of any changes made to the product or assessment of alternative products by the Standards and Assessments Section (in accordance with CE Electric UK Management Process PRC/200/203)
- A business or safety brief
- A training requirement
- A remedial or replacement works programme
- Disciplinary action where appropriate
- An ISO 9001: 2008 quality assurance audit on the supplier by the Standards and Assessments Section (in accordance with CE Electric UK Management Process PRC/200/204)

Actions shall be recorded on the defect summary recommendation paper and tracked by the Defect Process Co-ordinator via the data base.

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References

4.1 External Documentation

Reference	Title
ISO 9001: 2008	Quality Management system - Requirements

4.2 Internal documentation

Reference	Title
PRC/200/204	Management Process for the Audit of Suppliers Quality Management Systems
HAS/011	Accident Reporting and Investigations
HRP/005	Code of Practice and Business Ethics.

5. Definitions

Term	Definition
Low Voltage (LV)	A voltage up to and including 1000V
High Voltage (HV)	A voltage greater than 1000V, but less than 33, 000V
Extra High Voltage (EHV)	A voltage at 33, 000V and above
IRIS	Incident Reporting Information System
DINs	Dangerous Incident Notification
SOPs	Suspension of Operational Practice
NEDeRS	National Defect Reporting System
Defect Process	Monitors compliance with the agreed procedures
Co-ordinator (DPC)	
Defect Investigation	Manages specific defect investigations
Co-ordinator (DIC)	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Steven Salkeld	Standards Engineer	Steven Salkeld	25/04/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Andy Ellam	Network Performance and Risk Manager	Andy Ellam	29/04/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Jim Paine	Protection and Technical Services Manager	Jim Paine	07/05/2009
Chris Holdsworth	Standards and assessments Manager	Chris Holdsworth	26/03/2009

6.4 CDS Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
Sean Johnson	CDS Administrator	Sean Johnson	26/05/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
Mark Nicholson	Head of System Strategy	Mark Nicholson	29/04/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Mark Drye	Director of Asset Management	Mark Drye	08/05/2009

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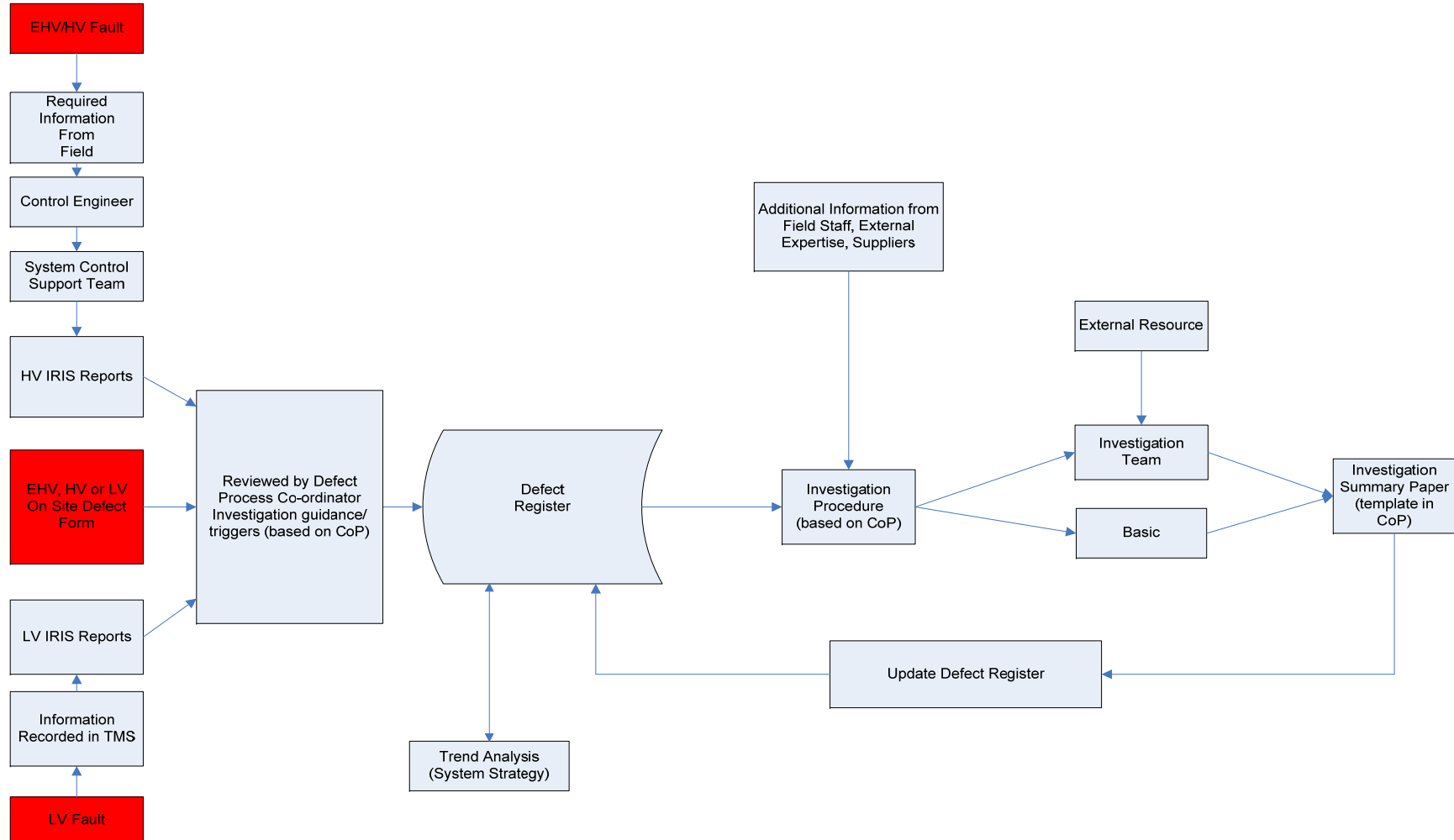
Appendix 1 – Defect Report Form

DEFECT
REPORTED BY: -

NAME:		Date:	
Designation:			
Contact Number:		Mobex/ Mobile	
Department and Location:			
Defective Item: (description, voltage etc.)			
Manufacturer:			
Commodity Code and/or Serial Number:			
Details of Defect / Abnormality: (Please provide as much background information as available, TMS fault number and attach photographs if necessary)			
Location of Defective Item: (Please detail any relevant hazards i.e. oil contamination etc)			

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Appendix 2 – Process Flow Chart



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Appendix 3 – Defect Summary Paper

Defect Reference Number
D _____

DEFECT INVESTIGATION

SUMMARY PAPER

FOR

Product Type or Reference

Commodity Code:

Authorisation:

Name	Title	Action	Signature	Date
	Investigation coordinator	Author		
		Review		
		Authorise		

Distribution:

Name	Title
Sean Caskey	Head of Programme Delivery
Mark Nicholson	Head of System Strategy
Steve Crosland	Head of Field Standards
Paul Norton	Head of SHEQ
Rod Gardner	Head of Network Operations
Mick Hickling	Head of Network Repairs
Geoff Bolden	Purchasing Operations Manager
	Originator

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(Title and/or Reference)

1. Summary of the Defect Investigation

Brief summary of the product, failure mode and scope of investigation.

2. Background

This is to include historical information which is generated from information supplied from others.

3. Defect Investigation

This is to include a brief summary of the investigation undertaken, including statements from external sources if applicable.

4. Conclusions

Provide a brief conclusion as to the cause of failure including any contributory factors such as environment, installation techniques and operational activities or manufacturing problems

5. Cost of Defect

*Provide brief explanation and quote direct costs which arise from external invoices, TMS or other sources.
Also include additional estimated costs incurred which may not be identifiable, such as time utilised by investigation team, other CE staff or others.
Where remedial programmes are proposed, identify potential future costs.*

6. Recommendations

This is to include, but not be limited to, statements on the following:

*Specification amendments
Installation amendments
Operational restrictions
Supplier QA requirements
Logistical issues*

7. Documented Actions

Required Action	Responsible	Due Date	Completed Date

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Appendix 4 – Defect Event and Action Log

Defect Investigation Co-ordinator	Defect	Defect Ref.

Action or Event	Person Responsible	Recorded Date	Action Complete Date

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Appendix 5 – Products to be Included

The following products are to be included (but not limited to) inclusion with the defect investigation process. Any product not detailed must firstly be agreed with the Defect Investigation Process Co-ordinator before an investigation can be initiated.

Underground System Associated Products

- Cable including - 132kV, 66kV, 33kV, 20kV, 11kV, LV mains and service, pilot and telephone, fuse tails, meter tails and earthing cable.
- Cable joints and terminations for the cables detailed above.
- Cable jointing sundries including tapes, solders, lugs etc.
- Cable ducts and protective tapes.

Overhead Line Associated Products

- Wood poles and foundation baulks.
- Steel poles.
- Conductor including AAAC, ACSR, PVC covered copper and aluminium, bare copper and aluminium, LV ABC and steel stay wire.
- Overhead line switchgear including – ABSD's, PMAR, expulsion switches and electronic sectionalisers.
- Surge arresters.
- Steelwork including – steel sections and channels, conductor fittings, insulator fittings and stay support accessories.
- Insulators including – post, tension, suspension, pin, stay and triggered spark gap type.
- Preformed helical fittings.
- Conductor compression and mechanical fittings.
- Barbed wire.

Plant and Switchgear

- Transformers including - system, CER, voltage, current, pole mounted distribution, ground mounted distribution and voltage regulators.
- Circuit breakers.
- Fault Throwers.
- Arc suppression coils.
- Neutral earthing resistors.
- Fault passage indicators.
- Switchgear remote control systems.
- Batteries and battery systems.
- Insulating oil.
- LV fuse boards including – distribution, feeder pillar and linkbox types.

LV Mains and Service Equipment

- Industrial service units.
- Wall and pole mounted outdoor boxes.
- Meter cupboards.

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- Service cut outs including – house service, heavy duty, street lighting and pole mounted.

Miscellaneous

- Earth rods, tape and fittings.
- Pad locks and door furniture.
- Safety signs and notices.
- Fasteners and fixings.
- HV and LV fuses.
- Cable cleats and guards.
- Protection equipment
- Telecoms equipment